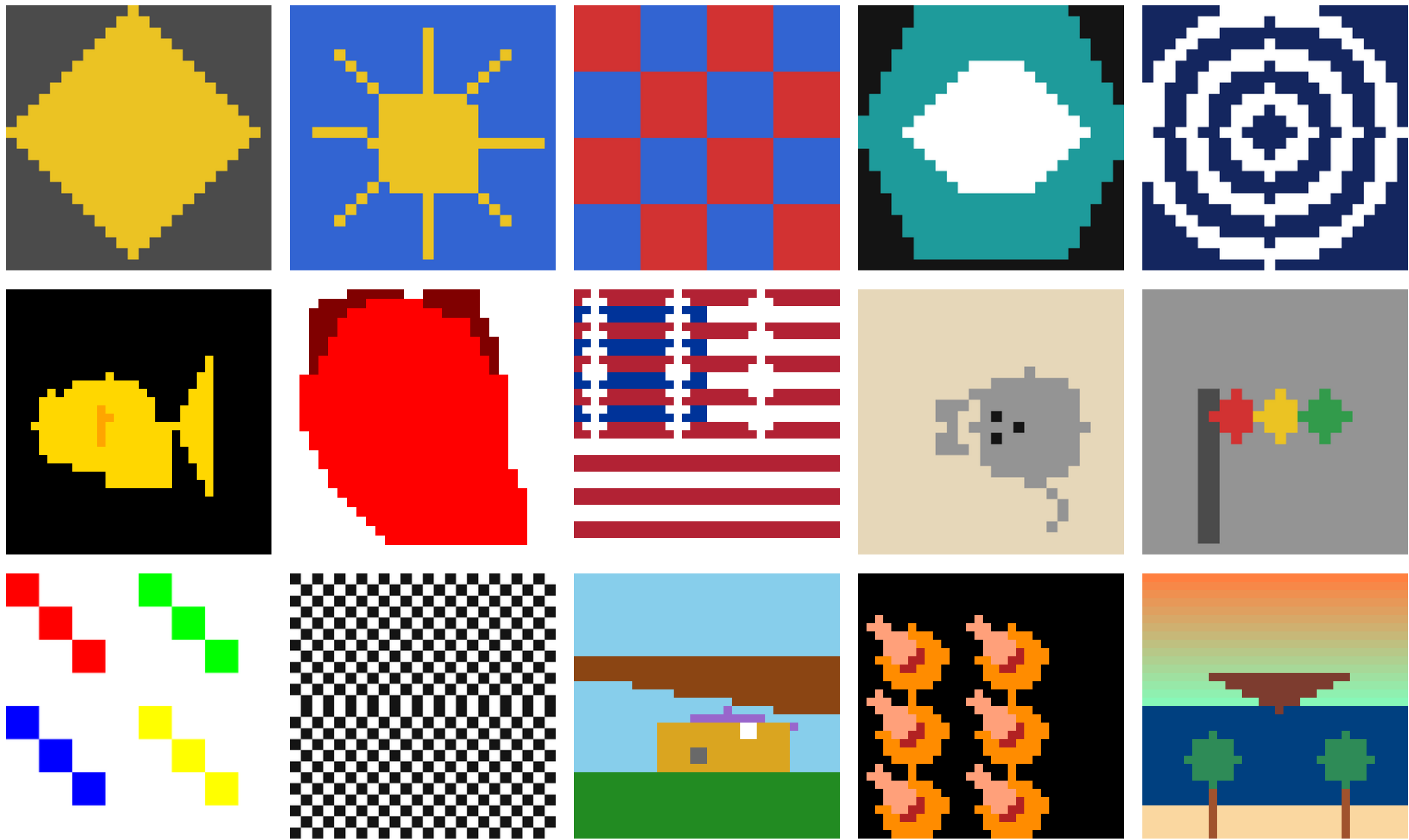


Autoregressive Mosaics



Probing 2D spatial reasoning in text-only language models

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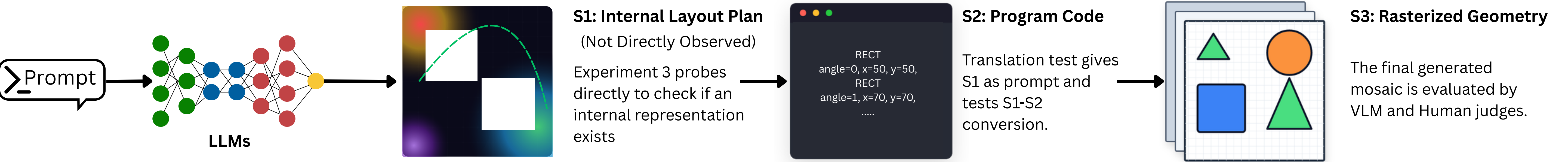
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RESEARCH QUESTION

To what extent can text-only language models compose and reason about 2D spatial layouts, how does the output medium limit this capability, and does the model construct a useful internal spatial representation, or does it primarily learn to express spatial descriptions through code?

THE BENCHMARK

One canvas. Two different tests.



Pipeline from the paper (Fig. 1, upper strip). The internal layout is not directly observed; probes test what can be decoded.

01 Translate a given layout

Every shape and coordinate is specified.
Can the model express them as code?
Scored by exact-area per-part IoU.

02 Compose a layout

The prompt leaves the arrangement open.
Can the model decide what goes where?
Scored by two VLM judges (0–5).

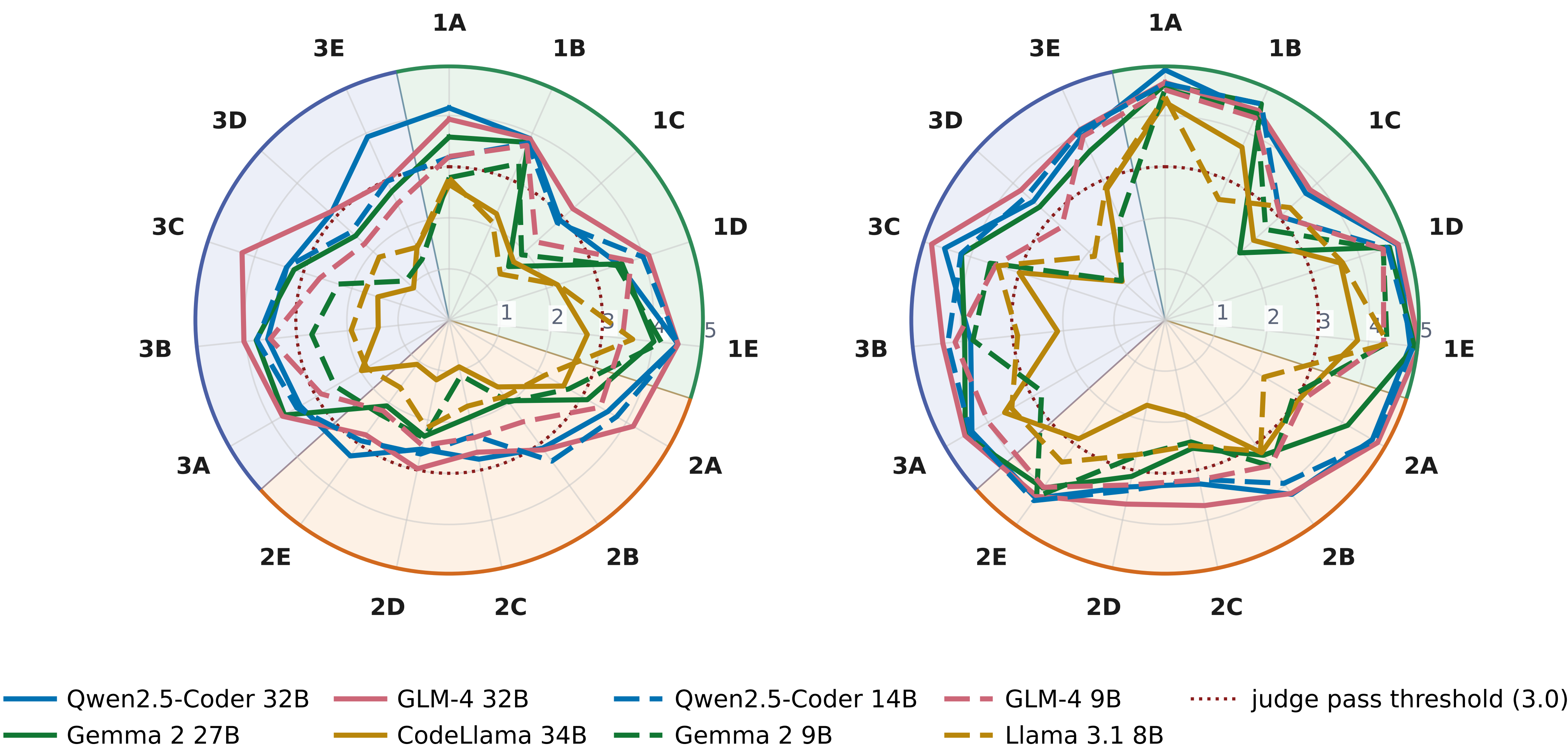
8 text-and-code models / 150 layout prompts / 24 × 24 canvas / 13,200 attempts

THE RESULTS

Models differ. Both judges agree.

(a) Judge 1 (Qwen2.5-VL-7B)

(b) Judge 2 (InternVL3-8B)



Reading the radars Higher is better; score range 0–5.

1A–E: elemental 2A–E: iconic 3A–E: compositional.
Both judges give the same model-level ordering ($\rho = 1.0$).
Color partitions (1E) are easiest; objects (2C) and scenes (3D) are hardest.

The model matters. So does the medium.

Scope. Eight models (8B–34B), coarse 2D images, judge-dependent scores. Detailed shared/specific probes and code-coordinate interventions cover two models; these do not establish a causal internal drawing plan.

THREE KEY FINDINGS

1.0

Translation is reliable.

Median PloU for every model; all clear the 0.6 gate. Composition still varies.

+0.37

The medium matters.

SVG improves the pooled layout score at matched resolution.
95% CI: [0.26, 0.47].

+0.164

A coarse layout is decodable.

Before generation: mean ΔR^2 above a text baseline. The specific drawing plan remains unproven.

WHEN A SHAPE MOVES More drawing follows.

Continuation rate: unchanged → moved
GLM-4 32B 43.8% → 84.6%
Gemma 2 27B 2.5% → 41.2%

Moved into unused space; completed outputs.
Textual coordinate edits · two models only.